

Medical Image Segmentation Using Machine Learning & Deep Learning



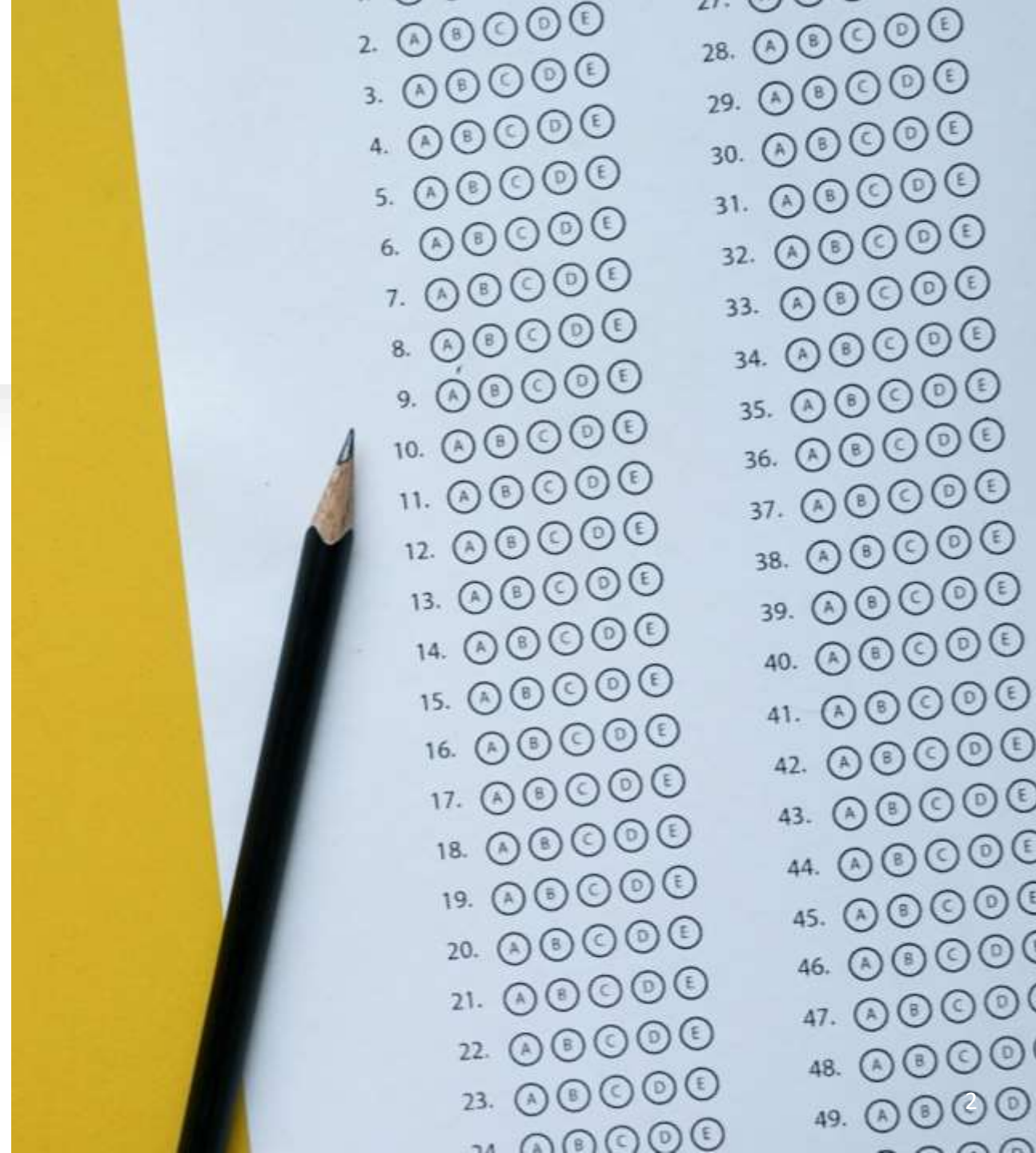
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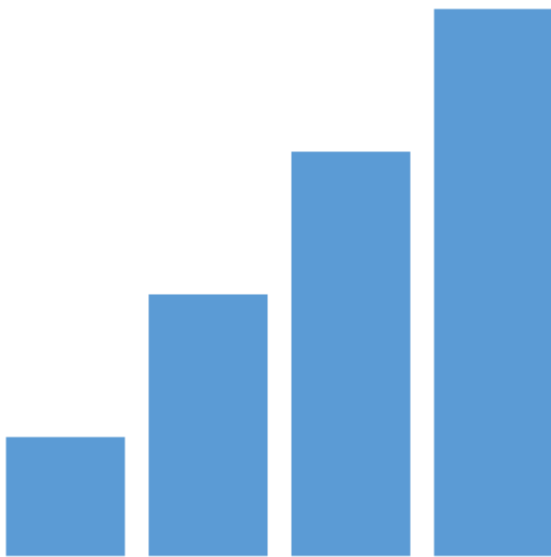
1. Introduction

- **Overview of Medical Image Segmentation:** Medical image segmentation is the critical process of partitioning a digital medical image (such as MRI, CT scans, X-rays, or Ultrasound) into multiple segments or regions of interest (ROIs), such as organs, tissues, or pathological lesions (e.g., tumors).
- **Role in Modern Healthcare:** Accurate segmentation is a foundational step in computer-aided diagnosis (CAD), surgical planning, radiation therapy, and longitudinal disease monitoring.
- **Evolution of Techniques:** Traditional segmentation relied heavily on manual delineation (prone to high inter-observer variability and time consumption) or basic algorithmic methods (thresholding, region growing). Modern approaches leverage advanced Machine Learning (ML) and Deep Learning (DL) to automate and enhance precision.



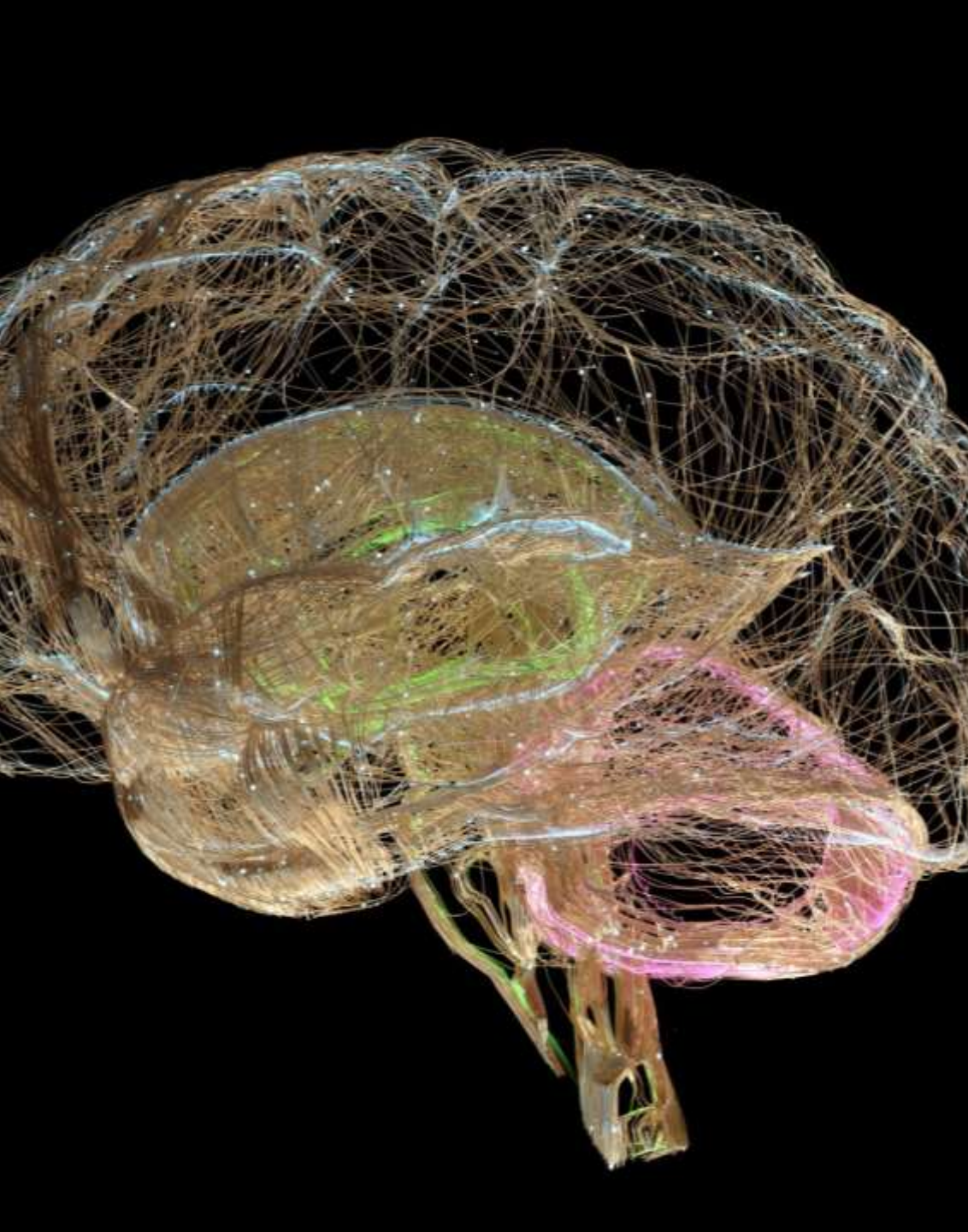
1.1. Motivation

- **Overcoming Manual Limitations:** Manual segmentation by radiologists is extremely time-consuming, labor-intensive, and subjective, often varying significantly across clinicians depending on experience.
- **Clinical Precision & Speed:** Automated systems provide rapid, reproducible, and highly accurate quantitative analysis, assisting clinicians in making faster and more reliable diagnostic decisions.
- **Improving Patient Outcomes:** Early and precise detection of anomalies (like micro-tumors or early-stage lesions) directly translates to timely interventions and improved survival rates.
- **Technological Advancement:** The rapid evolution of neural network architectures (like U-Net and its variants) provides powerful feature-extraction capabilities capable of handling complex anatomical variations.



1.2. Challenges

- **High Inter- and Intra-Patient Variability:** Anatomical structures vary widely in shape, size, and position across different patients, and even across different scans of the same patient.
- **Complex Boundaries and Low Contrast:** Soft tissue boundaries are frequently blurry, noisy, or suffer from low contrast gradients (e.g., in MRI/CT scans), making edge detection extremely difficult.
- **Pathological Heterogeneity:** Tumors and lesions often exhibit irregular textures, varying intensities, and diffuse margins, blurring the distinction between healthy tissue and pathology.
- **Class Imbalance:** In many datasets, the region of interest (e.g., a small lesion) occupies only a tiny fraction of the overall image pixels/voxels, causing models to bias heavily toward the background.



2. Problem Statement

- **Brain Tumor Segmentation:** We implement a Random Forest model to accurately identify tumor boundaries in MRI scans & CT scans. This approach focuses on efficient feature extraction to achieve high precision, specifically optimized for smaller, curated datasets.
- **Chronic Wound Analysis:** We utilize a deep learning model U-Net (Encoder + Decoder) structure & Rest 50 Net to segment the wounded region. These segmented masks are used to quantitatively analyze the wound area and track healing progress, providing a comprehensive diagnostic tool.



3. Literature Survey

- **Traditional Machine Learning & Random Forests:** Early automated segmentation pipelines relied heavily on traditional machine learning classifiers like Random Forest decision tree ensembles and Support Vector Machines (SVM). These methods used hand-crafted features (such as intensity gradients and texture descriptors) combined with spatial filters to differentiate tumor tissues from normal brain matter. While computationally efficient, they struggled with complex anatomical variability and diffuse tumor borders.
- **The Deep Learning Paradigm:** The emergence of Convolutional Neural Networks (CNNs) revolutionized medical image processing by bypassing manual feature engineering, automatically learning a hierarchy of discriminative features directly from raw MRI scans.
- **State-of-the-Art Volumetric Architectures:** Modern research heavily utilizes advanced encoder-decoder architectures, particularly the **3D U-Net** and its multi-modal variants (e.g., Attention U-Net), which capture volumetric spatial context across multi-sequence MRI data (T1, T1c, T2, FLAIR) to delineate heterogeneous tumor sub-regions like edema and enhancing cores.
- **Research Gap:** Despite high accuracy benchmarks, existing literature notes persistent bottlenecks in multi-modal generalization, boundary ambiguity between healthy and pathological tissue, and extreme class imbalance where small tumor cores occupy minimal voxel space

4. Methodology

- Overview of the end-to-end medical image processing, machine learning, and deep learning pipelines.
- Data acquisition framework and multi-class categorical structure for brain tumor scans.
- Automated image pre-processing, noise-reduction protocols, and normalization techniques.
- Feature engineering and spatial matrix transformation for classical machine learning models.
- Encoder-decoder architecture selection and design for deep learning-based wound segmentation.
- Target label encoding, data augmentation, and dataset partitioning strategies.
- Model configuration spanning Random Forest classifiers and deep neural networks.
- Performance evaluation, pixel-level overlap metrics, and confusion matrix error analysis.

Methodology: Brain Tumour Classification Pipeline using Machine Learning

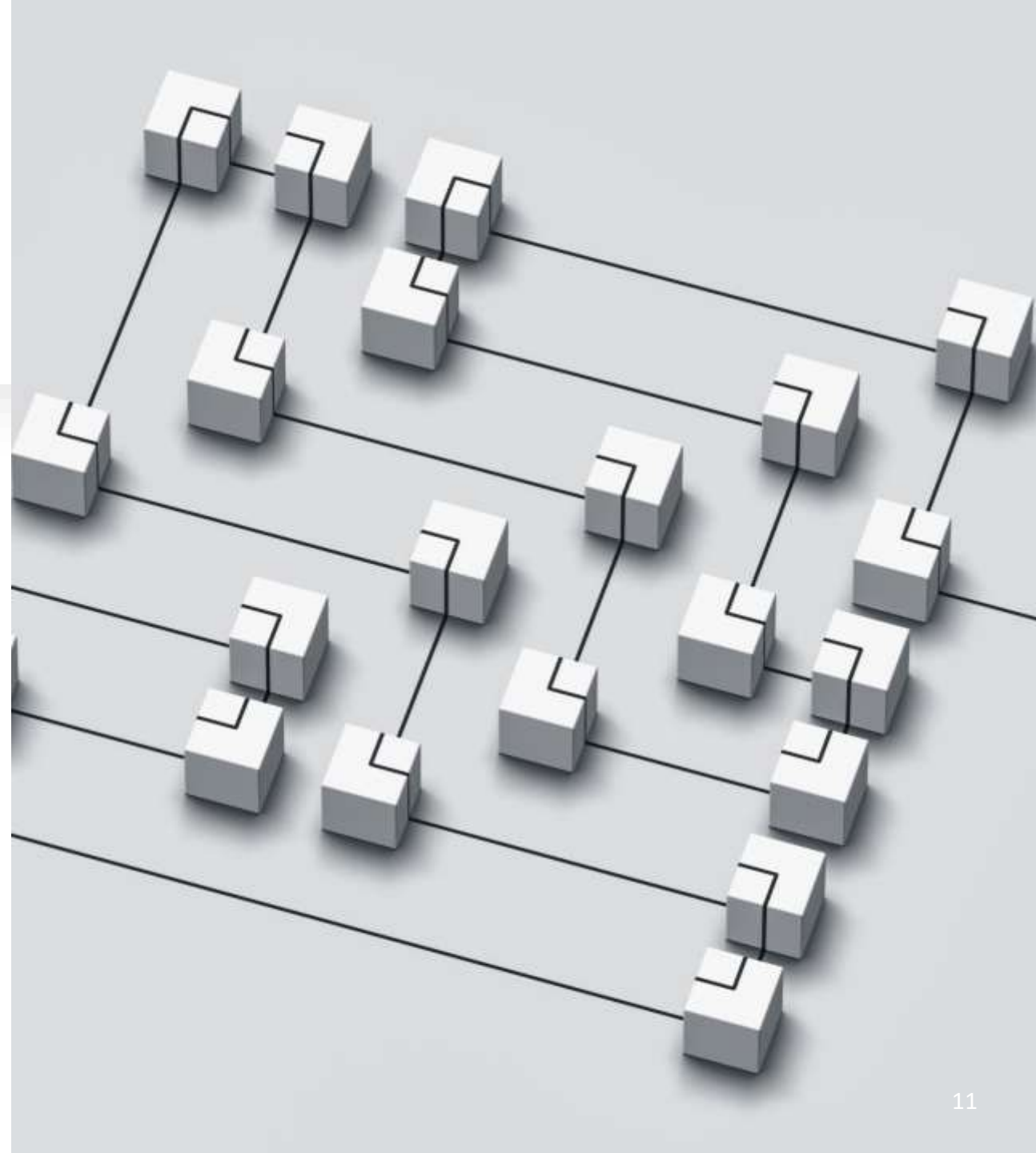
Process	Tools	Timeline
Data Acquisition	OS / Kaggle dataset	Loading structured multi-class MRI image directories.
Pre-processing	OpenCV (cv2)	Grayscale conversion, resizing to 64x64, median blur noise removal.
Feature Engineering	NumPy	Flattening 2D matrices into 1D numerical feature vectors.
Model Configuration	Scikit-learn	Random Forest classifier with 100 estimators (random state=42).

Methodology: Chronic Wound Analysis Pipeline using Deep Learning

Pipeline Stage	Tool / Framework	Operation Description
Data Pre-processing	OpenCV / NumPy	Resizing wound images & masks to 256x256, normalization.
U-Net Architecture	TensorFlow / Keras	Encoder-decoder network with skip connections for spatial preservation.
ResNet50 Backbone	Keras Applications	Integrated pre-trained encoder layers for enhanced feature extraction.
Evaluation Metrics	Scikit-learn / Metrics	Dice Coefficient, IoU Score, Precision, Recall, and F1-Score.

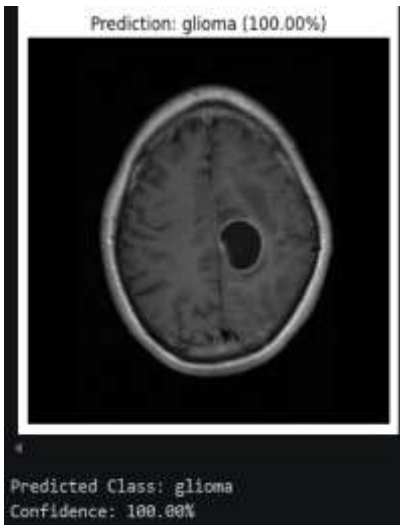
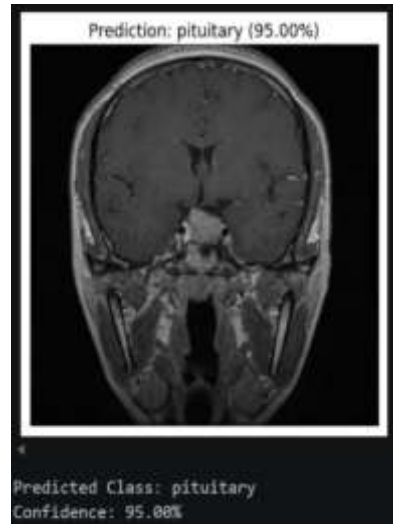
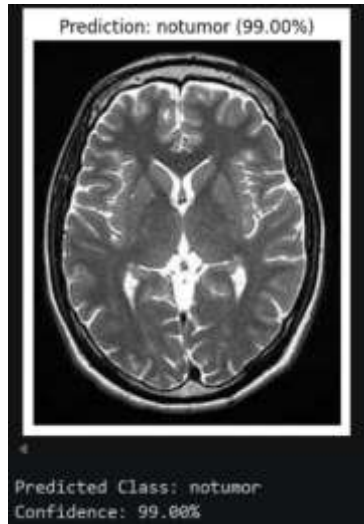
5. Experimental Results and Analysis

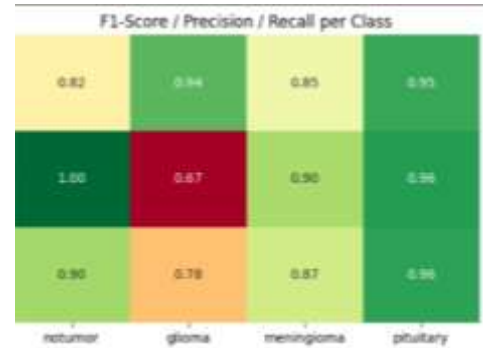
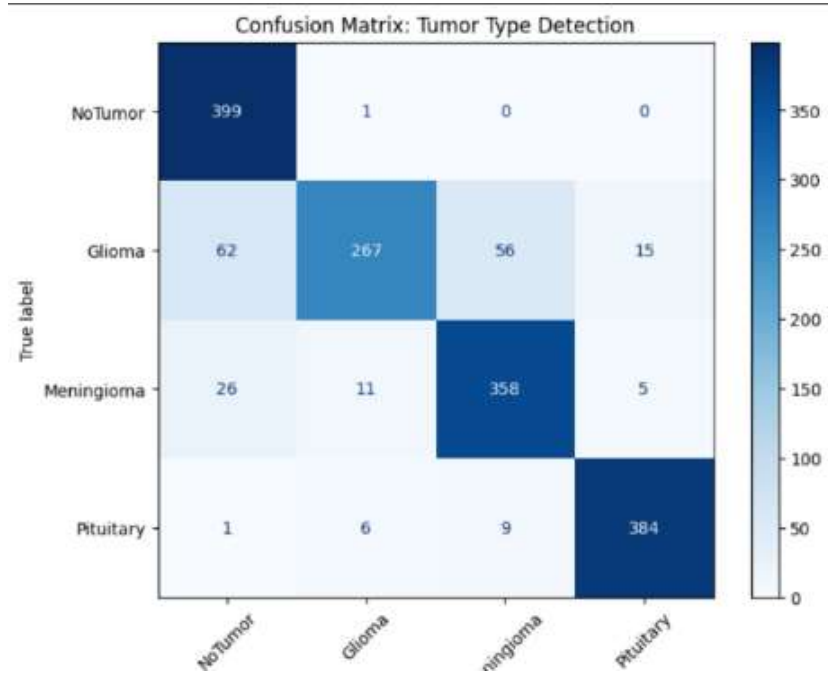
- Overview of the evaluation framework for both machine learning and deep learning models.
- Assessment of brain tumor classification using precision, recall, and confusion matrix error analysis.
- Quantitative evaluation of wound segmentation accuracy using pixel-level overlap metrics.
- Comparative analysis of traditional classifier performance versus deep neural network architectures.



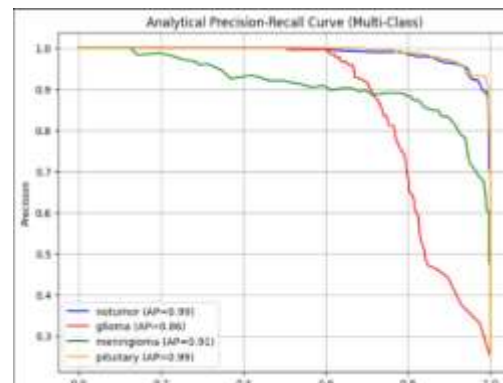
Brain Tumour Classification using Machine Learning

- The evaluation results indicate that the pipeline excels at identifying healthy brain scans and Pituitary cases with high precision. However, misclassifications within the Glioma category highlight boundary ambiguities that require refined feature extraction. Future optimizations will focus on incorporating advanced attention layers and data augmentation to improve overall diagnostic safety and accuracy.





Results & Analysis



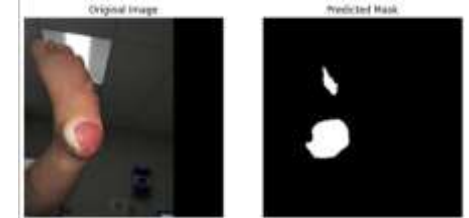
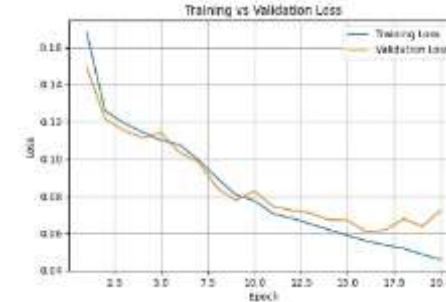
Classification Report:

	precision	recall	f1-score
notumor	0.82	1.00	0.90
glioma	0.94	0.67	0.78
meningioma	0.85	0.90	0.87
pituitary	0.95	0.96	0.96
accuracy			0.88
macro avg	0.89	0.88	0.88
weighted avg	0.89	0.88	0.88

Chronic Wound Analysis using Deep Learning

U-net Model

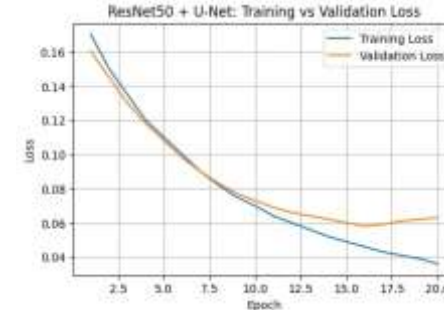
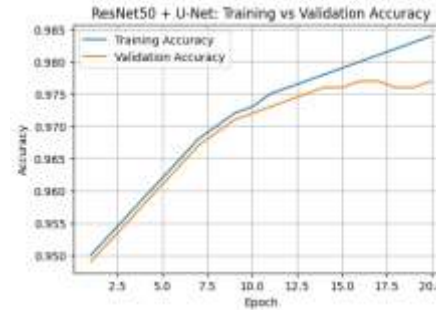
- The U-Net model successfully segmented the wound region from the input images. The training and validation accuracy increased steadily, while the loss decreased over 20 epochs, indicating effective learning.



Accuracy	0.97
Test Loss	0.06
Dice Coefficient	0.74
IOU Score	0.59
Precision	0.84
Recall	0.67
F1 Score	0.74

Rest 50 net Model

- The ResNet50 U-Net model achieved better feature extraction by using the ResNet50 encoder. It produced accurate segmentation results with high accuracy and improved Dice and IoU scores. Overall, the model showed robust performance and was effective for wound image segmentation.



Accuracy	0.97
Test Loss	0.36
Dice Coefficient	0.76
IOU Score	0.60
Precision	0.86
Recall	0.67
F1 Score	0.75

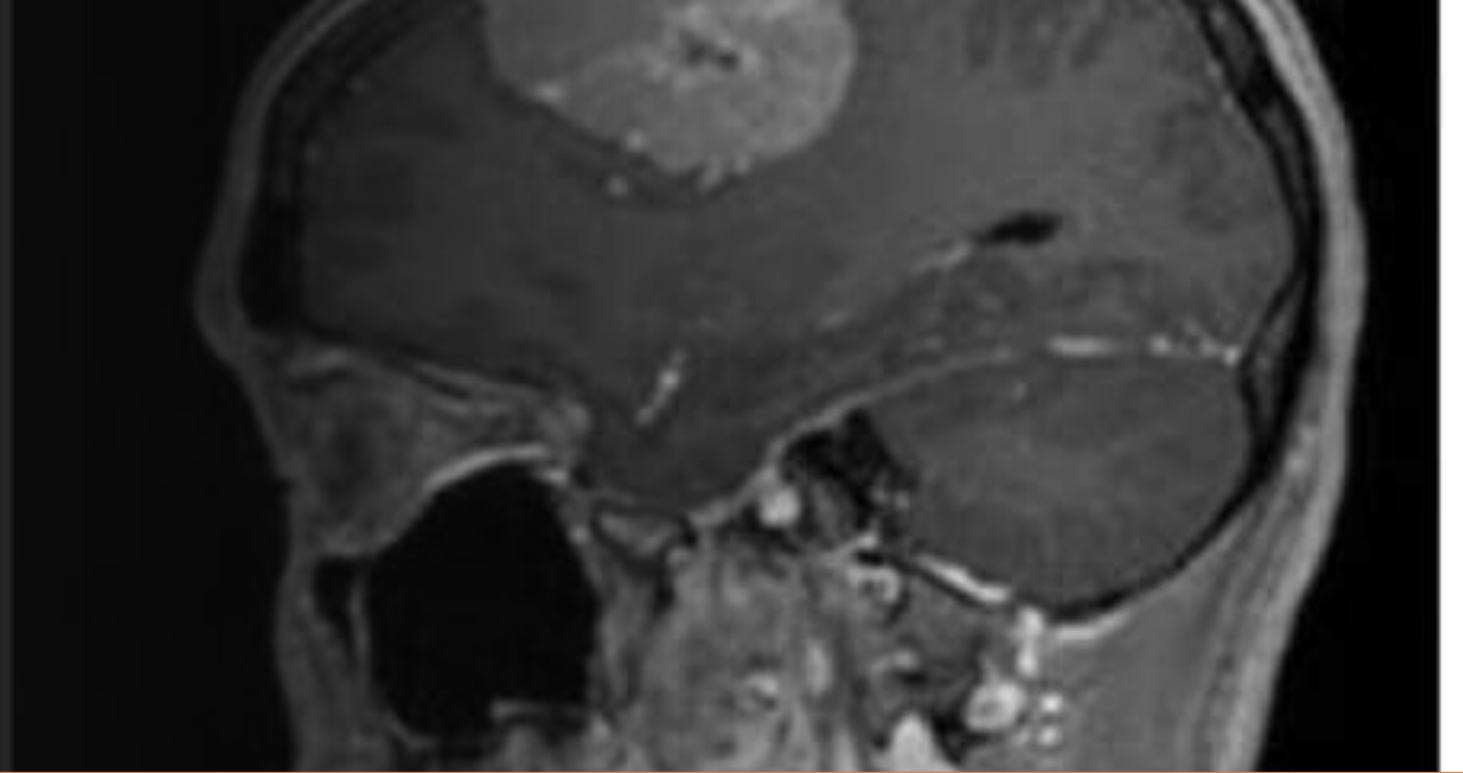
6. Conclusion and future work

- This research successfully designed, implemented, and evaluated two robust computational pipelines for medical image analysis. The Random Forest classifier demonstrated high efficacy in detecting pituitary tumours and healthy brain scans, while identifying crucial boundary ambiguities within glioma cases. Concurrently, the deep learning U-Net and ResNet50 U-Net architectures successfully captured intricate wound boundaries, showing steady loss reduction and strong spatial correlation through Dice and IOU evaluations. Together, these systems establish a reliable foundation for computer-aided clinical diagnostic tools.
- Future improvements and expansion of this research will focus on:
 - **Advanced Architectural Enhancements:** Integrating transformer-based blocks into both the classification and segmentation networks to capture finer global and local contextual features.
 - **Dataset Expansion and Augmentation:** Incorporating larger, multi-centric clinical datasets with diverse noise profiles to enhance model generalizability and prevent overfitting.
 - **Explainable AI and Clinical Deployment:** Implementing visual interpretation tools like Grad-CAM for the tumour classifier and deploying the wound segmentation model into an interactive, real-time clinical web application.

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Thank You



Original Image



Predicted Mask

